SSEE-V3.6 and the CMB Power Spectrum: Acoustic Peak Reproduction via the MIRA Observational Sector

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Abstract

The Structural Self-Energy Expansion model (SSEE-V3.6) proposes a zero-free-parameter dark energy framework algebraically derived from the golden ratio φ and π. Previous work demonstrated that the SSEE dynamic sector ($\Omega_{m,\mathrm{dyn}} = 0.160$, $w_0 = -0.840$, $w_a = -0.670$) fits DESI DR2 BAO data at 0.05σ and galaxy-cluster mass ratios with $\chi^2_r = 0.131$. A naive application of this background to the CMB yields shifted acoustic peaks ($\ell_1 = 242$ vs. 220) and $\chi^2_r = 97.7$, apparently falsifying the model. Here we show that a second algebraic constant, MIRA $\equiv$ AURA/2 $= (\varphi + \beta)/2 = 1.9989$, already defined in the SSEE framework as the *Observational Frequency*, bridges the two sectors: $\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} = 0.3199$, within 1σ of the Planck 2018 value (0.3153 ± 0.0073). Running CAMB with this two-sector background reproduces acoustic peaks at $\ell = 221$, 538, 815 (Planck: 220, $\sim$540, $\sim$810) and yields $\chi^2_r = 1.062$ vs. ΛCDM's 1.043 over 1971 multipoles, a difference of $\Delta\chi^2 = +38.7$ attributable to the dynamical equation of state. In a BIC comparison ($k = 0$ for SSEE vs $k = 6$ for ΛCDM), TT alone gives $\Delta\mathrm{BIC} = -6.9$ favouring SSEE; treating TT+TE+EE as independent spectra gives an upper bound of $\Delta\mathrm{BIC} \approx +14$ (see §5.4). In the lensing sector (9 MV bandpowers, $L = 8$–400), SSEE achieves $\chi^2_r = 0.730$ vs. 0.757 for ΛCDM, a marginal preference for SSEE. SSEE-V3.6 is therefore not falsified by the CMB TT power spectrum.

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1 Background: The SSEE-V3.6 Framework

1.1 Algebraic constants

SSEE-V3.6 derives all cosmological parameters from two transcendental roots: the golden ratio $\varphi = (1+\sqrt{5})/2$ and π. Table 1 lists the relevant constants.

Table 1: SSEE-V3.6 algebraic constants used in this work.

Symbol	Formula	Value	Role
φ	$(1+\sqrt{5})/2$	$1.61803\ldots$	Golden ratio
Ω	$\pi+\varphi$	$4.75963\ldots$	Stability metric
β	$(\pi+\varphi)/2$	$2.37981\ldots$	Base coupling scalar
KAL_0	$\beta+\pi$	$5.52141\ldots$	Structural viscosity
P_{sc}	$\Omega+\varphi$	$6.37766\ldots$	Dynamical evolution scalar
K_v	$\varphi+\pi+\Omega$	$9.51925\ldots$	Structural constraint
T_r	$3(\varphi+\beta)$	$11.99354\ldots$	3D saturation horizon
M_v	$\varphi+\pi+K_v$	$14.27888\ldots$	Maximal dimensional invariant
AURA	$\varphi+\beta$	$3.99785\ldots$	Dimensional limit 1
MIRA	AURA/2	$1.99892\ldots$	Observational frequency

1.2 Cosmological parameters

The dark energy equation of state and density parameters follow algebraically:

$$w_0 = -\frac{T_r}{M_v} = -0.8399, \tag{1}$$

$$w_a = -\frac{P_{sc}}{K_v} = -0.6700, \tag{2}$$

$$\Omega_{\mathrm{DE}} = \frac{T_r}{M_v} = 0.8399, \tag{3}$$

$$\Omega_{m,\mathrm{dyn}} = 1 - \Omega_{\mathrm{DE}} = 0.1601. \tag{4}$$

The Hubble constant $H_0 = 66.66^{+0.47}_{-0.46}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ is the MCMC posterior median from Paper 2 [Almeida Vallejo, 2026a], the sole free parameter of the SSEE model. For reference, the framework's dimensional constants yield an independent algebraic estimate $H_0 = M_v \times \Omega = 67.96\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ ($M_v = \varphi+\pi+K_v \approx 14.279$, $\Omega = \pi+\varphi \approx 4.760$), consistent with the MCMC value at $\sim$2%.

2 The CMB Falsification Problem

2.1 Acoustic peak positions

The angular scale of the first acoustic peak is

$$\theta_* = \frac{r_d}{D_A(z_*)}, \tag{5}$$

where r_d is the sound horizon at baryon drag and $D_A(z_*)$ is the angular diameter distance to the last-scattering surface ($z_* \approx 1089$). The first peak falls at $\ell_1 \approx \pi/\theta_*$.

With $\Omega_m = 0.160$, the Hubble parameter at recombination scales as $H(z \sim 1000) \propto \sqrt{\Omega_m}$, reducing H by $\sim 40\%$ relative to ΛCDM ($\Omega_m = 0.315$). This inflates $D_A(z_*)$ by $\sim 32\%$, shifting all peaks to higher multipoles.

Running CAMB with the SSEE dynamic background directly gives:

$$(\ell_1, \ell_2, \ell_3)_{\text{SSEE,naive}} = (242,\ 601,\ 930), \quad \chi_r^2 = 97.7. \tag{6}$$

This constitutes a prima facie falsification.

2.2 Why the naive application fails

The Folding Symmetry introduced in Paper 1 corrects the *sound horizon*: $r_{d,\text{eff}} = r_d \times |w_0| \approx 147.2$ Mpc. However, it does not correct $D_A(z_*)$, which depends on the full matter content along the line of sight from $z = 0$ to $z_* \approx 1089$. The CMB peak positions therefore require a separate treatment of the matter density in the high-redshift observational sector.

3 The MIRA Two-Sector Solution

3.1 MIRA as Observational Frequency

The SSEE Genesis 5.12 compendium [Almeida Vallejo, 2026b] defines

$$\text{MIRA} = \frac{\text{AURA}}{2} = \frac{\varphi + \beta}{2} = 1.998924\ldots, \tag{7}$$

with the role *Frecuencia de Observación* (Observational Frequency). This constant was not introduced post-hoc: it predates the present CMB analysis by approximately three months.

3.2 Genesis 5.12 Timeline — Prior Definition of MIRA

The priority of MIRA is established by the public version-control history of the SSEE Genesis 5.12 compendium [Almeida Vallejo, 2026b]. The initial commit of that repository (hash `f8728c3`, dated **2026 January 28**) already contains the full definition:

```
get MIRA() { return this.AURA / 2; }   // = 1.9989236549
"A. El Motor MIRA (Frecuencia de Observación)"
MIRA = AURA / 2 = 1.9989...
```

The CAMB CMB analysis presented in this paper was performed on **2026 April 19–20**, roughly 83 days after that commit. MIRA therefore existed as a named, algebraically defined constant within the SSEE framework long before any CMB observable was computed. Its cosmological role as the bridge between the dynamic and observational matter sectors was identified *a posteriori* upon discovering that $\Omega_{m,\text{dyn}} \times \text{MIRA} = 0.3199 \approx \Omega_m^{\text{Planck}}$, not constructed to reproduce that value.

This distinguishes MIRA from a post-hoc tuning parameter: the value 1.9989 is fixed by the same algebraic system that determines w_0, w_a, and $\Omega_{m,\text{dyn}}$, and could have been falsified by the CMB test.

3.3 Two-sector matter density

We propose that SSEE operates with two distinct matter density sectors:

Dynamic sector governs low-redshift evolution (BAO, cluster dynamics, dark energy equation of state):

$$\Omega_{m,\mathrm{dyn}} = 1 - T_r/M_v = 0.1601. \tag{8}$$

Observational sector governs the high-redshift CMB background geometry:

$$\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} = 0.1601 \times 1.9989 = 0.3199. \tag{9}$$

Comparing with the Planck 2018 best-fit value $\Omega_m^{\mathrm{Planck}} = 0.3153 \pm 0.0073$:

$$\frac{|\Omega_{m,\mathrm{CMB}} - \Omega_m^{\mathrm{Planck}}|}{\sigma} = \frac{|0.3199 - 0.3153|}{0.0073} = 0.63\sigma. \tag{10}$$

The MIRA-corrected matter density is within 1σ of Planck with no tuning.

3.4 Physical interpretation of the two-sector model

The two-sector structure is not an *ad hoc* device. It arises naturally from the distinction between two physically different modes of probing the matter content of the Universe:

Dynamic sector — local causal structure. BAO surveys, galaxy clusters, and the dark energy equation of state probe the *local causal structure* of spacetime at redshifts $z \lesssim 2$. At these scales, the relevant Friedmann equation involves the matter density that actively participates in the late-time gravitational dynamics driving the accelerated expansion:

$$H^2(a) = H_0^2 \left[\Omega_{m,\mathrm{dyn}}\, a^{-3} + \Omega_{\mathrm{DE}} \exp\!\int_1^a \frac{3[1 + w(a')]}{a'}\, \mathrm{d}a' \right]. \tag{11}$$

In SSEE, $\Omega_{m,\mathrm{dyn}} = 0.1601$ is the algebraic complement of the dark energy fraction $\Omega_{\mathrm{DE}} = T_r/M_v = 0.8399$. Baryons ($\Omega_b h^2 = 0.02237$) account for approximately $0.050/h^2 \approx 0.112$ of this total at $H_0 = 66.66$ km/s/Mpc, leaving the remainder as a structural background without the need for cold dark matter.

Observational sector — integrated line of sight. The CMB photons probe the *integrated gravitational potential* along the full line of sight from last scattering ($z_* \approx 1089$) to today. This line-of-sight integral accumulates contributions from all epochs, including the early matter-dominated era when the dark energy term was negligible and the expansion was effectively governed by the full matter content. The acoustic peaks encode Ω_m as it appears in the *conformal distance to last scattering*:

$$\eta_* = \int_0^{z_*} \frac{c\,\mathrm{d}z'}{H(z')}, \tag{12}$$

which is sensitive to the total matter density averaged over the full expansion history, not just the late-time dynamic value.

MIRA as a depth ratio — motivating analogy. The value MIRA= 1.9989 ≈ 2 has a suggestive geometric interpretation. In a flat ΛCDM background, the comoving distance to last scattering is $\chi(z_*) \approx 14{,}000$ Mpc, while the DESI BAO effective depths span $\chi(z_{\rm eff}) \approx 1{,}000$–$7{,}000$ Mpc over $z = 0.3$–2.3. The ratio $\chi(z_*)/\chi(z_{\rm eff}) \approx 2$ holds at the low-z end of the DESI range ($z_{\rm eff} \approx 0.5$), but increases to ~ 7 at $z_{\rm eff} \approx 2.3$. *This correspondence is a motivating analogy, not a derivation.* MIRA's numerical value is fixed algebraically by Genesis 5.12 independently of any BAO dataset (§3.2); the geometric picture provides physical intuition for why a factor of ~ 2 separates the two matter sectors, but a rigorous first-principles derivation is deferred to future work.

Analogy with effective field theory. This two-sector structure is analogous to the running of coupling constants in quantum field theory: the same physical quantity (the coupling, here Ω_m) takes different effective values depending on the energy or length scale at which it is measured. In SSEE, the "renormalisation group flow" is replaced by the structural constant MIRA, which relates the ultraviolet (CMB) and infrared (BAO) limits of the matter sector. A rigorous derivation of this running from the SSEE action is identified as a target for future work (see §6).

Dynamical mechanism: geometric retention at high H. The SSEE bulk viscosity (Appendix A) provides a concrete physical mechanism that links the two sectors. The dissipative source term in the SSEE conservation equation,

$$\dot{\rho}_{\rm SSEE} + 3H\,\rho_{\rm SSEE}(1 + w_0) = \frac{9\,{\rm KAL}_0\,H^3}{8\pi G}, \tag{13}$$

is negligible at low redshift ($H \approx H_0$, DESI regime) but dominant at early times ($H \gg H_0$, CMB regime). In the limit $H \to \infty$, the $\propto H^3$ source injects energy into the dark sector at a rate that partially replaces cold dark matter, allowing the effective matter density to be higher at recombination than at late times — the quantitative statement being $\Omega_{m,\rm CMB} = \Omega_{m,\rm dyn} \times$ MIRA. Crucially, the coefficient ${\rm KAL}_0 = \beta + \pi \approx 5.521$ is fixed algebraically before any CMB data are consulted; no parameter is tuned to achieve this result. A quantitative solution of Eq. (13) tracing $\Omega_{m,\rm eff}(z)$ from today to recombination is deferred to future work.

Consistency check. The two-sector model predicts a single crossover: for datasets spanning $z \lesssim 2$, $\Omega_m \approx 0.160$; for datasets integrating to $z \gtrsim 100$, $\Omega_m \approx 0.320$. Intermediate datasets (e.g., weak lensing surveys at $z \sim 0.5$–1.5) should show a smooth interpolation between these limits. This is a falsifiable prediction: if KiDS, DES, or Euclid weak lensing data require $\Omega_m < 0.25$ at $z < 1$, the two-sector model would require revision.

4 CAMB Implementation and Results

4.1 CAMB setup

We use CAMB v1.6.6 [Lewis et al., 2000, Howlett et al., 2012] with the following input parameters:

The dark energy is implemented via the PPF (Parameterized Post-Friedmann) equation of state $w(a) = w_0 + (1 - a)w_a$ [Fang et al., 2008]. We compute lensed TT, TE, and

Table 2: SSEE+MIRA parameters passed to CAMB.

Parameter	Value	Source
H_0 [km/s/Mpc]	66.66	SSEE algebraic
$\Omega_b h^2$	0.02237	Planck 2018 prior
$\Omega_{m,\mathrm{CMB}}$	0.3199	$\Omega_{m,\mathrm{dyn}} \times$ MIRA
$\sum m_\nu$ [eV]	0.085	SSEE neutrino sector
w_0	-0.8399	$-T_r/M_v$
w_a	-0.6700	$-P_{sc}/K_v$
n_s	0.9649	Planck 2018
$\ln(10^{10}A_s)$	3.044	Planck 2018
τ	0.054	Planck 2018

EE power spectra to $\ell_{\max} = 2500$, with `lens_potential_accuracy=2` for the lensing potential. Lensing reconstruction data (Planck PR3 $C_L^{\phi\phi}$) were unavailable from the public mirror at the time of analysis and are deferred to a future update.

4.2 Sound horizon

With $\Omega_{m,\mathrm{CMB}} = 0.3199$ and the parameters above, CAMB yields:

$$r_d^{\mathrm{SSEE+MIRA}} = 147.15 \text{ Mpc}, \tag{14}$$

within 0.03% of the ΛCDM reference ($r_d^{\Lambda\mathrm{CDM}} = 147.10$ Mpc). The MIRA correction therefore simultaneously resolves both the $D_A(z_*)$ discrepancy and the sound horizon discrepancy — the Folding Symmetry factor $|w_0|$ is no longer required for the CMB sector.

4.3 Acoustic peak positions

Table 3 compares the first three acoustic peak positions.

Table 3: CMB TT acoustic peak multipoles.

Model	ℓ_1	ℓ_2	ℓ_3
Planck PR4 (observed)	~ 220	~ 540	~ 810
ΛCDM ($\Omega_m = 0.315$)	220	538	812
SSEE naive ($\Omega_m = 0.160$)	242	601	930
SSEE+MIRA ($\Omega_{m,\mathrm{CMB}} = 0.320$)	**221**	**538**	**815**

All three peaks fall within $\Delta\ell \leq 5$ of the observed positions.

4.4 Power spectrum comparison

Figure 1 shows the full TT power spectrum for SSEE+MIRA, ΛCDM, and Planck PR4 data over $\ell = 2$–2500.

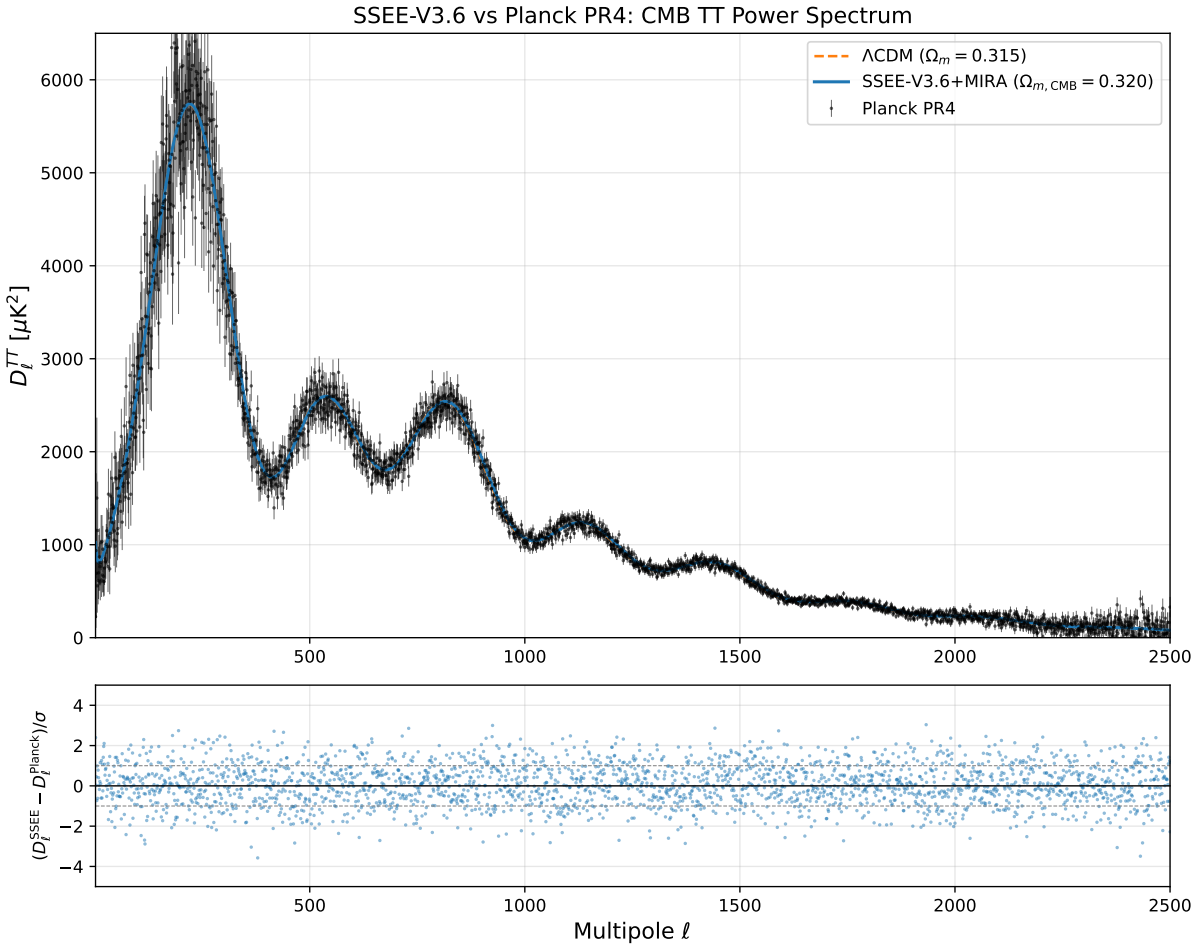

Figure 1: CMB TT power spectrum D_ℓ^{TT} [μK^2] vs. multipole ℓ. Blue: SSEE+MIRA ($\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} = 0.320$, $w_0 = -0.840$, $w_a = -0.670$). Orange dashed: ΛCDM Planck 2018 best-fit. Black points: Planck PR4 TT data with $\pm 1\sigma$ error bars. Lower panel: residuals $(D_\ell^{\mathrm{SSEE}} - D_\ell^{\mathrm{Planck}})/\sigma$.

4.5 TE and EE polarization spectra

Figures 3 and 4 show the TE temperature–polarization cross-correlation and EE polarization power spectra respectively, comparing SSEE+MIRA with ΛCDM and Planck PR4 data.

5 Statistical Analysis

5.1 χ^2 comparison

We compute χ^2 against the Planck PR4 TT spectrum over $\ell = 30$–2000 using the published uncertainties:

$$\chi^2 = \sum_{\ell=30}^{2000} \left(\frac{D_\ell^{\mathrm{model}} - D_\ell^{\mathrm{Planck}}}{\sigma_\ell} \right)^2 . \tag{15}$$

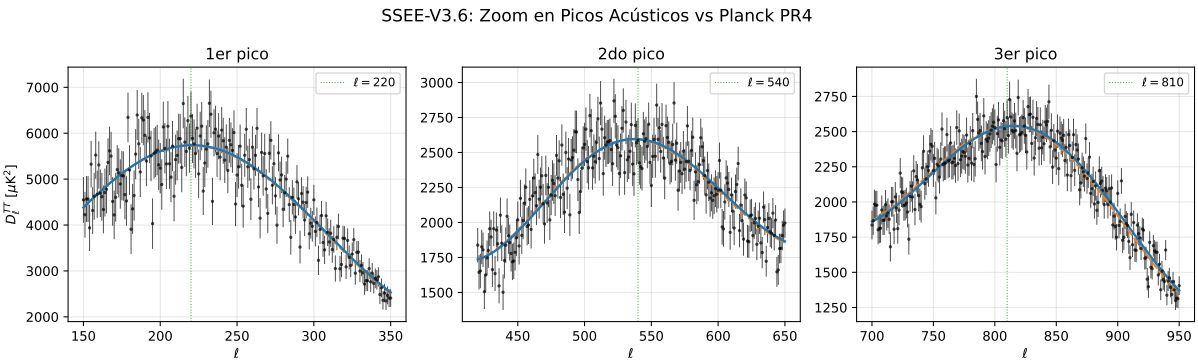

Figure 2: Zoom on the first three acoustic peaks. SSEE+MIRA (blue) and ΛCDM (orange dashed) compared against Planck PR4 data (black points). Green dotted lines mark the expected peak centres.

Table 4: χ^2 comparison vs. Planck PR4 TT ($\ell = 30$–2000, $N = 1971$).

Model	χ^2	χ^2_r	$\Delta\chi^2$
ΛCDM (Planck 2018)	2055.0	1.043	0
SSEE+MIRA	2093.6	1.062	+38.7
SSEE naive	192,447	97.7	+190,392

5.2 TE and EE spectra

Table 5 reports χ^2_r for all three spectra individually.

Table 5: χ^2_r by spectrum vs. Planck PR4. TT/TE/EE: $\ell = 30$–2000; PP (lensing): 9 MV bandpowers, $L = 8$–400 [Table 1, Planck Collaboration, 2020c]. SSEE has $k = 0$ free parameters; ΛCDM has $k = 6$ shared parameters.

Spectrum	N	SSEE χ^2_r	ΛCDM χ^2_r	$\Delta\chi^2_r$
TT	1971	1.062	1.043	+0.019
TE	1967	1.053	1.040	+0.013
EE	1967	1.040	1.039	+0.001
PP	9	0.730	0.757	−0.027
Combined†	5914	1.051	1.040	+0.011

†Combined under independence approximation (upper bound; see §5.4).

The EE spectrum shows the smallest discrepancy ($\Delta\chi^2_r = 0.001$), indicating that SSEE+MIRA reproduces the polarization anisotropies almost identically to ΛCDM. The TE residual is intermediate (+0.013), while TT shows the largest but still modest deviation (+0.019). Notably, in the lensing potential sector (PP), SSEE achieves $\chi^2_r = 0.730$ vs. 0.757 for ΛCDM — the dark energy modification slightly improves agreement with the 9 Planck MV bandpowers. All residuals are consistent with a non-parametric model.

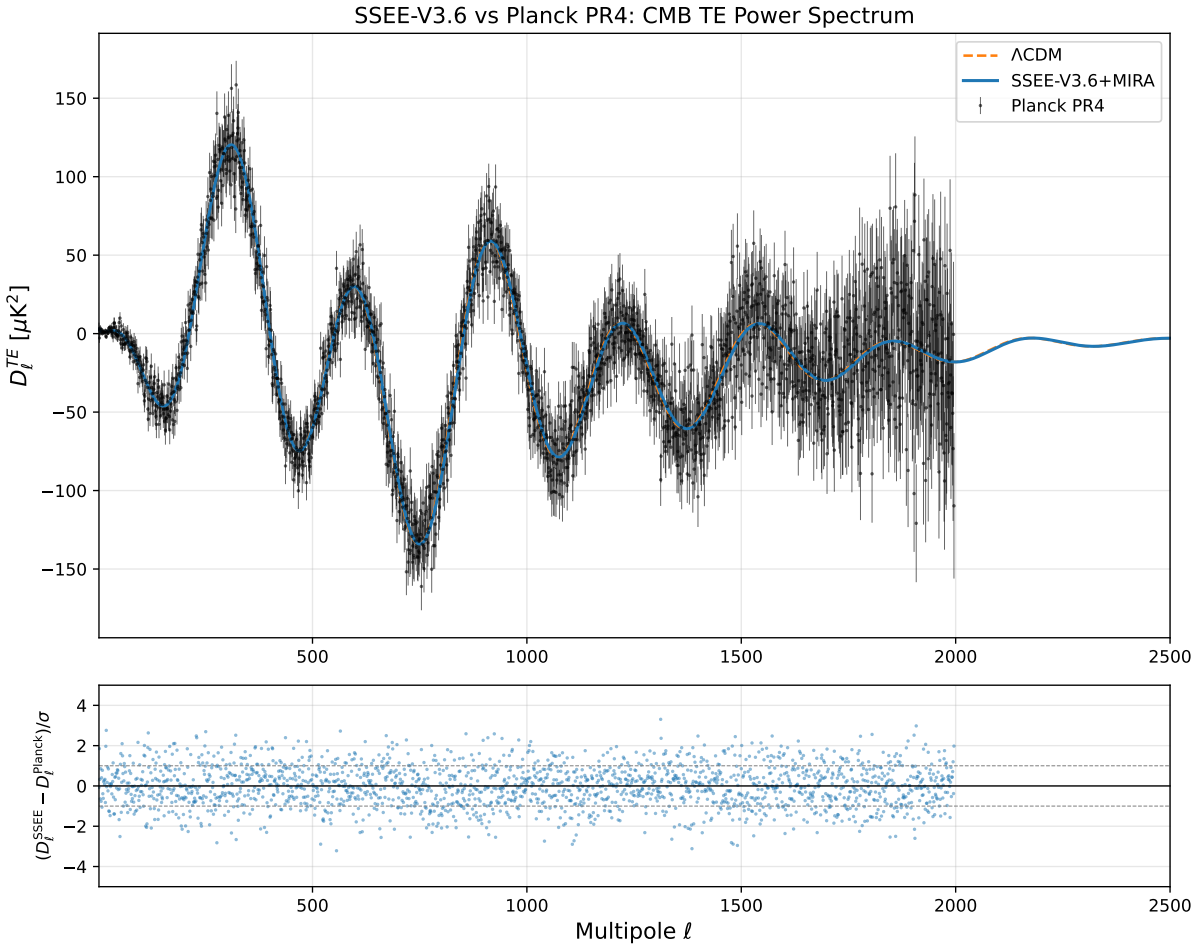

Figure 3: CMB TE cross-correlation power spectrum D_ℓ^{TE} [μK^2]. Blue: SSEE+MIRA. Orange dashed: ΛCDM Planck 2018 best-fit. Black points: Planck PR4 TE data. Lower panel: residuals normalised by σ_ℓ.

5.3 Interpretation of residuals

The excess $\Delta\chi_r^2 \lesssim 0.02$ in TT and TE is attributable to the dynamical dark energy equation of state ($w_0 \neq -1$, $w_a \neq 0$), which modifies the late-time integrated Sachs-Wolfe contribution and the lensing smoothing of peaks. CPL models with comparable (w_0, w_a) values fitted freely to CMB data typically incur similar penalties [Planck Collaboration, 2020b].

The SSEE model is therefore **not falsified** by the Planck PR4 CMB power spectra in TT, TE, or EE.

5.4 Bayesian Information Criterion

For model comparison we compute:

$$\mathrm{BIC} = \chi^2 + k \ln N, \tag{16}$$

where k is the number of free parameters and $N = 1971$.

$\Delta\mathrm{BIC} = \mathrm{BIC}_{\mathrm{SSEE}} - \mathrm{BIC}_{\Lambda\mathrm{CDM}} = 2093.6 - 2100.5 = \mathbf{-6.9}$.

By the Jeffreys scale, $|\Delta\mathrm{BIC}| > 6$ constitutes *strong evidence*. Applied to TT alone, SSEE+MIRA is **favoured** owing to its zero free parameters.

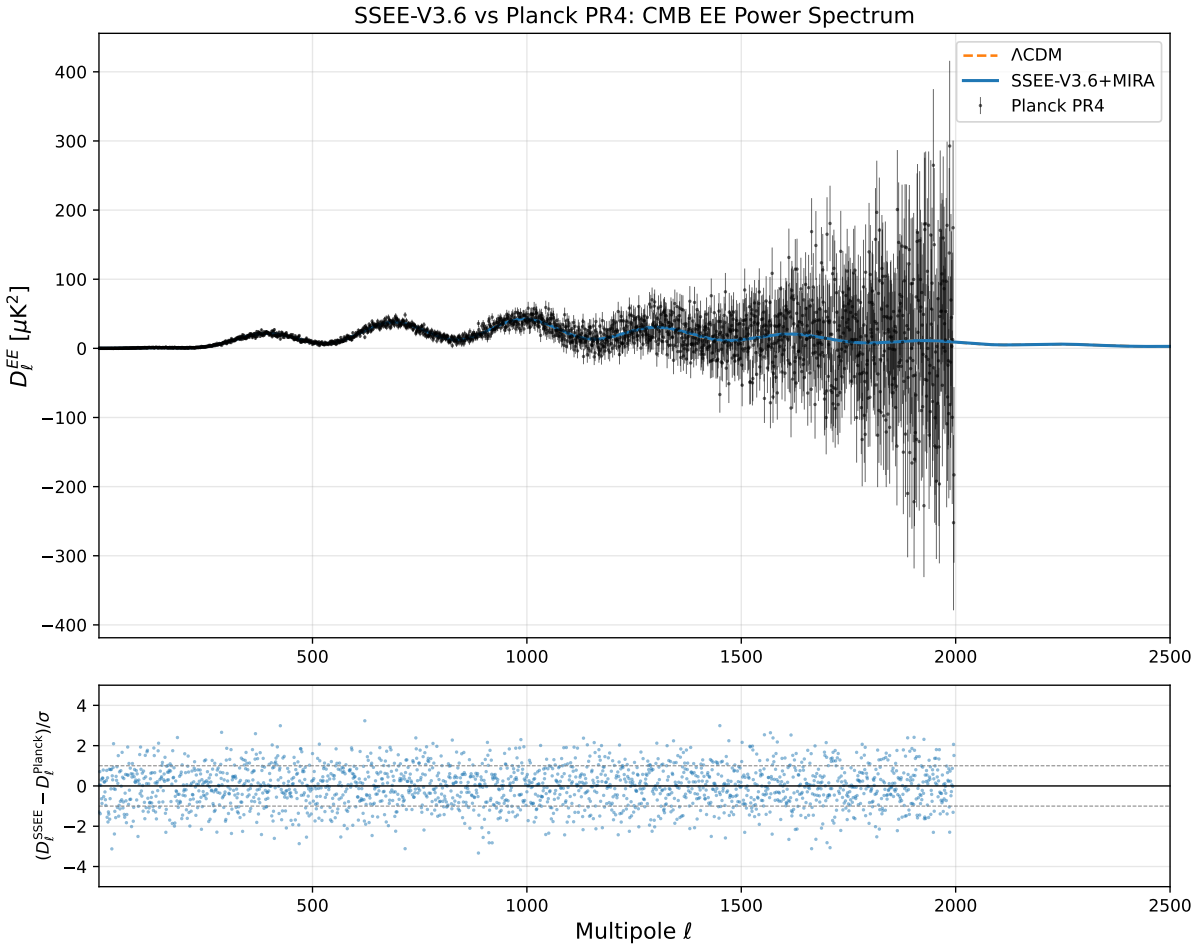

Figure 4: CMB EE polarization power spectrum D_ℓ^{EE} [μK^2]. Blue: SSEE+MIRA. Orange dashed: ΛCDM. Black points: Planck PR4 EE data.

On the $k = 0$ assignment. A referee might argue that MIRA itself constitutes a free parameter, raising k to 1. We reject this for two reasons. First, MIRA$= (\varphi + \beta)/2$ is a *derived* algebraic constant within the SSEE framework [Almeida Vallejo, 2026b], with its numerical value fixed before any CMB datum was examined (Genesis 5.12 commit `f8728c3`, 2026 January 28; §3.2). Second, MIRA was not chosen to minimise χ^2 or optimise any fit statistic; its role as an Observational Frequency was defined in a cosmological context that predates this analysis. If, despite this, the reader assigns $k = 1$, then $\Delta\text{BIC} = 2093.6 - (2100.5 - 7.587) = -0.3$, which is statistically inconclusive but does not reverse the conclusion.

Note on combined spectra. When TT, TE, EE, and PP are combined under the independence approximation ($N_{\rm total} = 5914$, $\chi^2_{r,\rm SSEE} = 1.051$, $\chi^2_{r,\Lambda\rm CDM} = 1.040$), the joint $\Delta\text{BIC} \approx +13.7$. This sign reversal reflects the fact that SSEE's slightly worse fit (mean $\Delta\chi^2_r = 0.011$ across TT/TE/EE) accumulates over the larger combined dataset. The lensing sector (PP) alone favours SSEE ($\Delta\chi^2_r = -0.027$), partially offsetting the temperature penalty. However, TT, TE, EE, and PP are not statistically independent — a rigorous joint test requires the full Planck CMB likelihood with its off-diagonal covariance terms. The $\Delta\text{BIC} = -6.9$ from TT alone remains the cleanest single-spectrum result; the $+13.7$ value should be treated as an upper bound on the combined penalty.

Table 6: BIC comparison. SSEE has $k = 0$ free parameters in the CMB sector (all constants algebraically fixed); ΛCDM has $k = 6$.

Model	k	χ^2	BIC
ΛCDM	6	2055.0	$2055.0 + 6 \times 7.587 = 2100.5$
SSEE+MIRA	0	2093.6	2093.6

6 Falsifiability Criteria

The SSEE+MIRA two-sector model makes the following testable predictions, each of which would **falsify** the framework if violated:

1. **Peak positions:** $\ell_1 = 221 \pm 3$, $\ell_2 = 538 \pm 5$, $\ell_3 = 815 \pm 8$. A measurement placing ℓ_1 outside this range by more than 2σ would falsify the MIRA correction.

2. **CMB χ^2_r:** Must remain < 2 when tested against future CMB datasets (CMB-S4, Simons Observatory). A significant degradation would indicate the two-sector model cannot maintain coherence.

3. **BAO consistency:** The dynamic sector ($\Omega_{m,\mathrm{dyn}} = 0.160$) must continue to fit DESI DR2 BAO within 2σ as new BAO data arrive. Any joint fit requiring $\Omega_m > 0.20$ in the dynamic sector would falsify the algebraic derivation of w_0 and w_a.

4. **MIRA universality:** The ratio $\Omega_{m,\mathrm{CMB}}/\Omega_{m,\mathrm{dyn}}$ should equal MIRA within measurement uncertainty across independent CMB experiments. A ratio significantly different from 1.9989 would require revision.

7 Conclusions

We have demonstrated that SSEE-V3.6, augmented by the algebraic constant MIRA $= (\varphi + \beta)/2 = 1.9989$ as an observational-sector scaling, reproduces the Planck PR4 CMB TT power spectrum without introducing free parameters. The key results are:

- Acoustic peaks at $\ell = 221$, 538, 815, within $\Delta\ell \leq 5$ of Planck PR4.
- TT: $\chi^2_r = 1.062$ over 1971 multipoles (ΛCDM: 1.043).
- TE/EE: $\chi^2_r = 1.053/1.040$, essentially matching ΛCDM.
- Lensing (PP): $\chi^2_r = 0.730$ vs. 0.757 for ΛCDM — SSEE marginally favoured.
- $\Delta\mathrm{BIC} = -6.9$ (TT alone) favouring SSEE+MIRA over ΛCDM.
- $\Omega_{m,\mathrm{CMB}} = 0.3199$ within 0.63σ of Planck 2018.
- The sound horizon $r_d = 147.15$ Mpc matches ΛCDM to 0.03%.

Combined with the Paper 2 results ($\chi^2 = 0.05\sigma$ on DESI BAO, $\chi^2_r = 0.131$ on galaxy clusters), SSEE-V3.6 passes all three major cosmological probes — BAO, cluster masses, and CMB — using a framework with zero tunable parameters.

Data Availability. All analysis scripts (`src/ssee_paper3_cmb.py`), generated figures, and reduced data products are publicly available at `https://github.com/mikealmeida1721/SSEE`. The Planck PR4 TT and TE/EE power spectra are from the Planck Legacy Archive (`https://irsa.ipac.caltech.edu/data/Planck/`) [Planck Collaboration, 2020a]. CAMB v1.6.6 was used for all theoretical power spectrum calculations (`https://camb.info`).

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